

Applied Deep Learning

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Gating Mechanisms

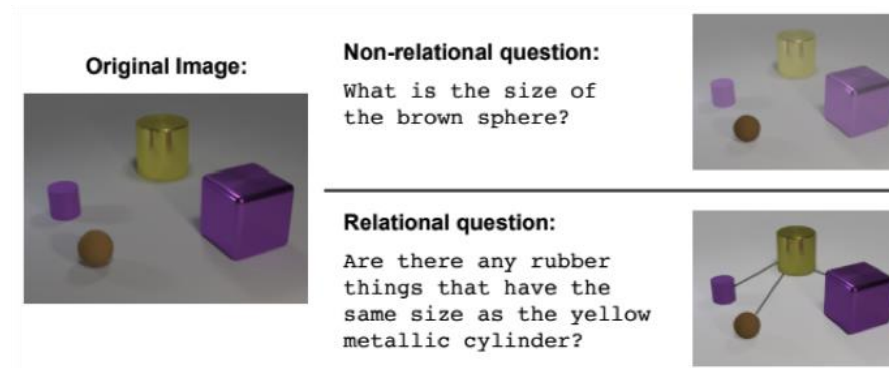


Gated Architectures Achievements

automatic
handwriting

Handwriting
Handwriting
Handwriting

visual reasoning



lip reading



Reminder of RNN and Derivatives

Model

$$h_t = \sigma(b + Wh_{t-1} + Vx_t)$$

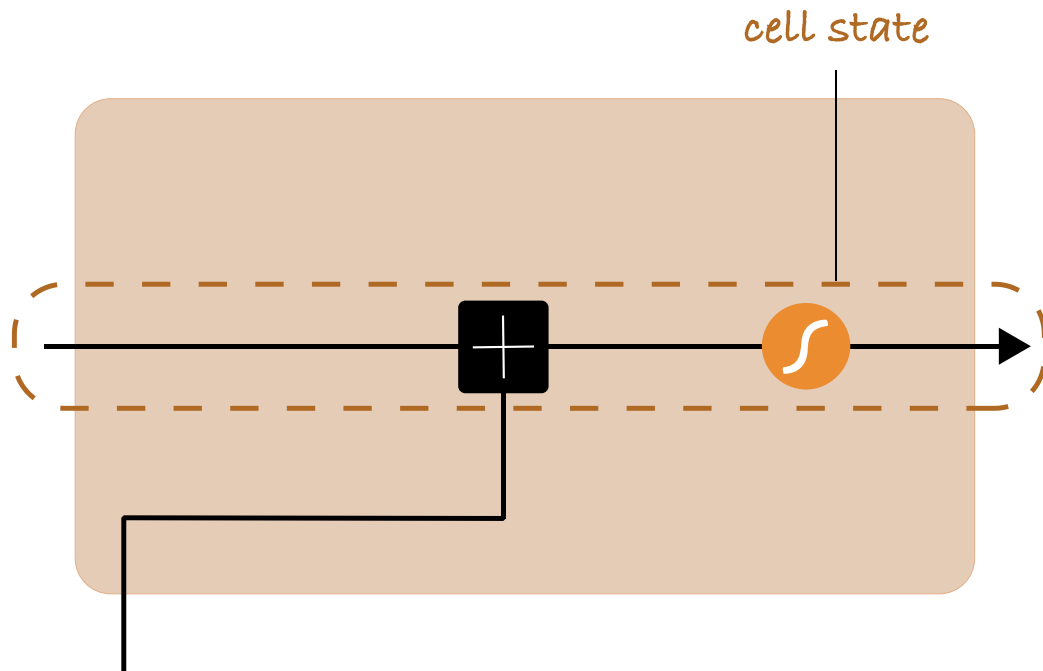
Derivative

$$\frac{\partial L_t}{\partial h_t} = \frac{\partial L_t}{\partial h_T} \prod_{k=t}^{T-1} \frac{\partial h_{k+1}}{\partial h_k} = \frac{\partial L_t}{\partial h_T} \prod_{k=t}^{T-1} D_{k+1} W_k^T$$

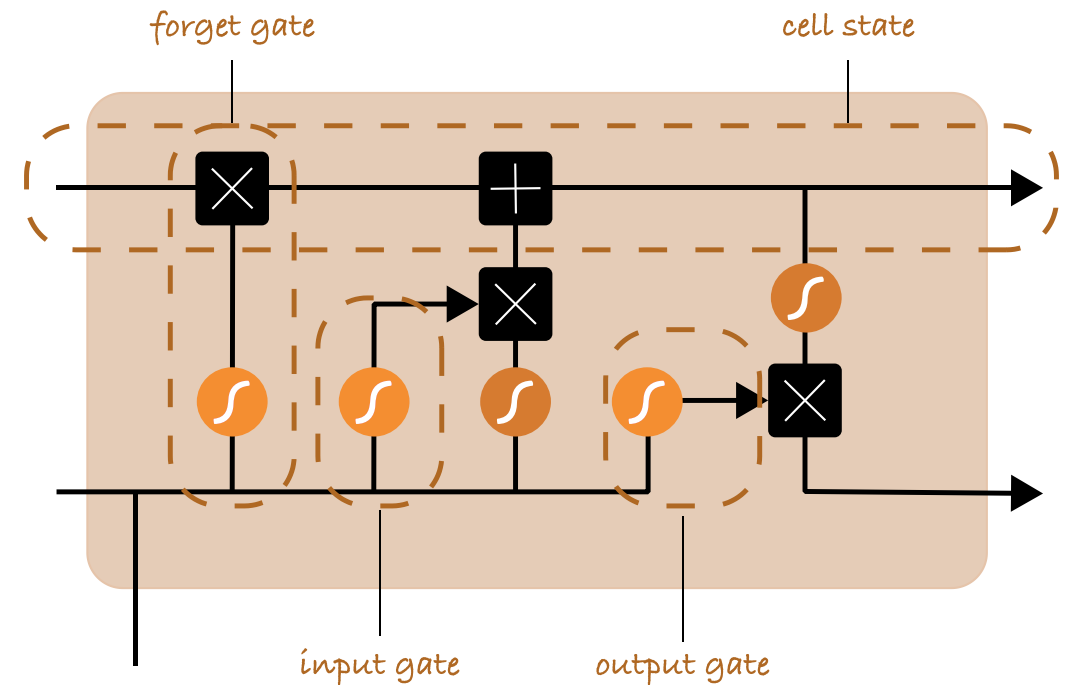


RNN vs. Long-Short Term Memory

RNN



LSTM



tanh



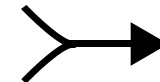
sigmoid



multiplication



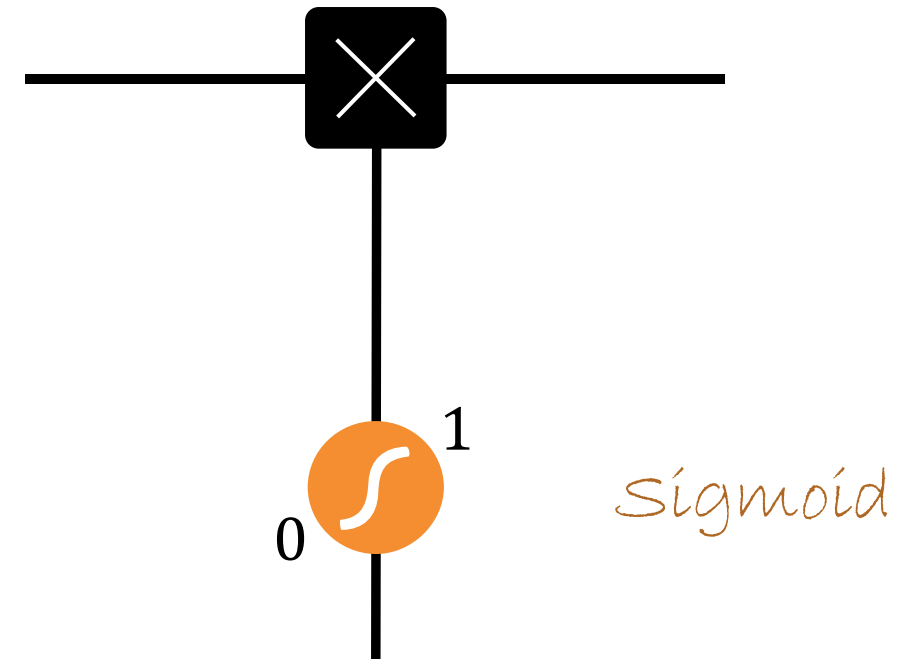
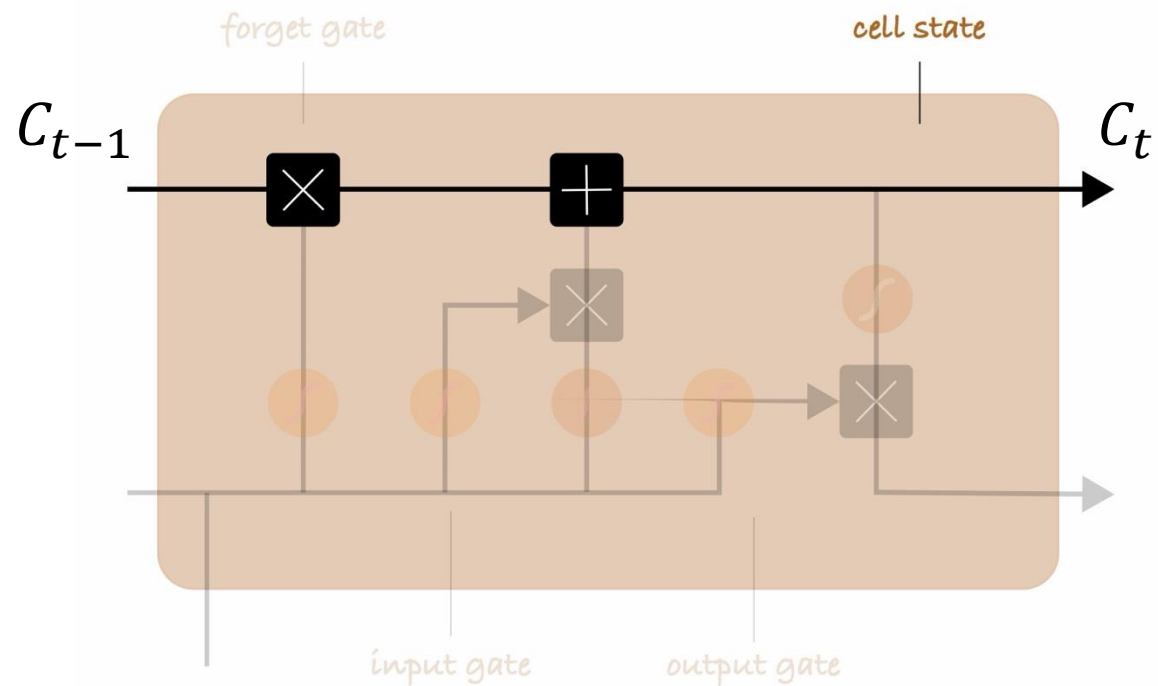
addition



concatenation



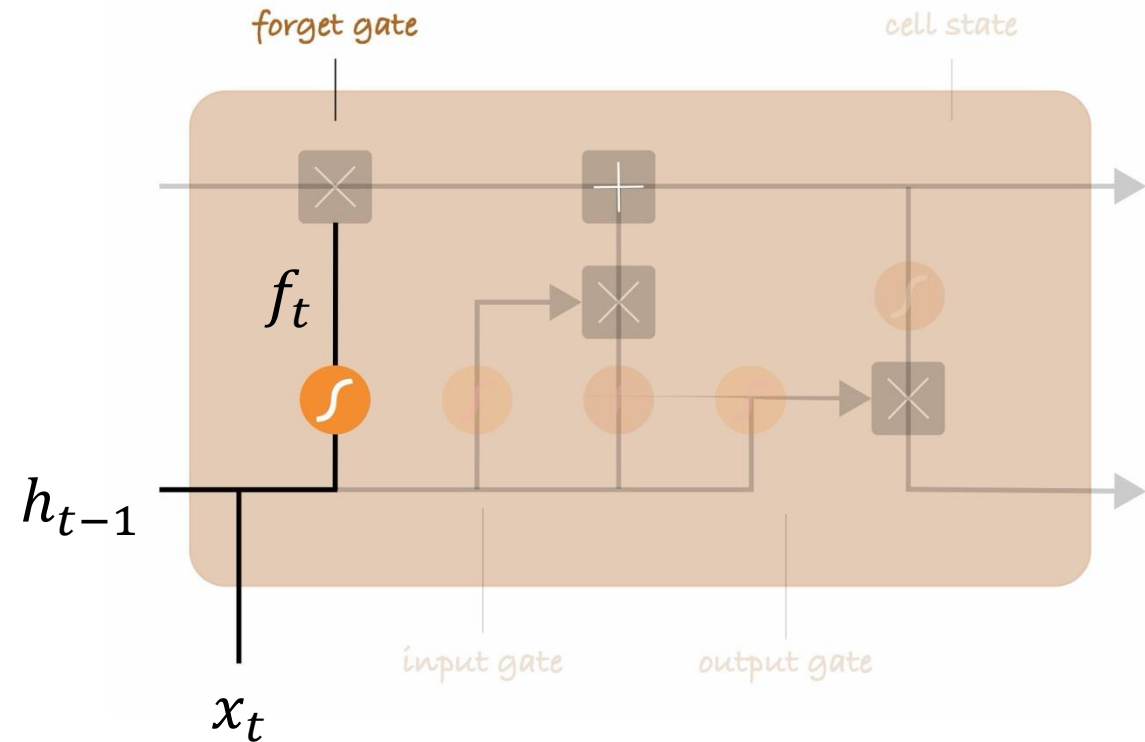
The Core Idea of LSTM



Step-by-Step LSTM Walkthrough

The **forget** gate layer

$$f_t = \sigma(W_f h_{t-1} + U_f x_t)$$

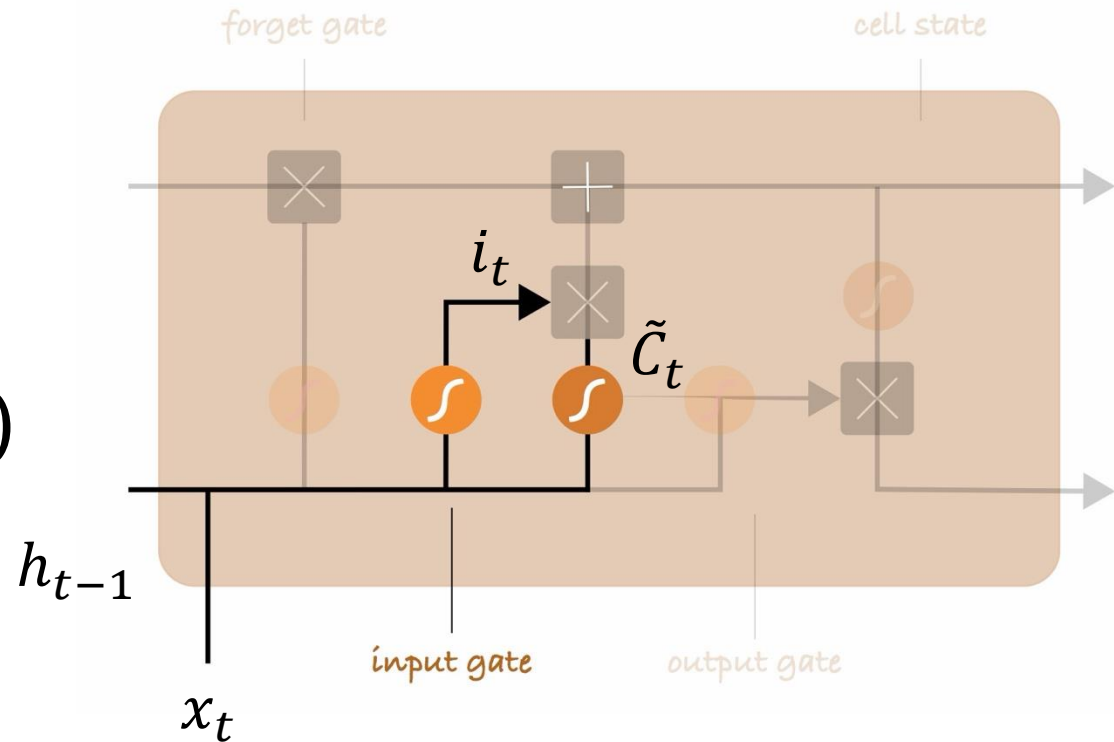


Step-by-Step LSTM Walkthrough

The **input** gate layer

$$i_t = \sigma(W_i h_{t-1} + U_i x_t)$$

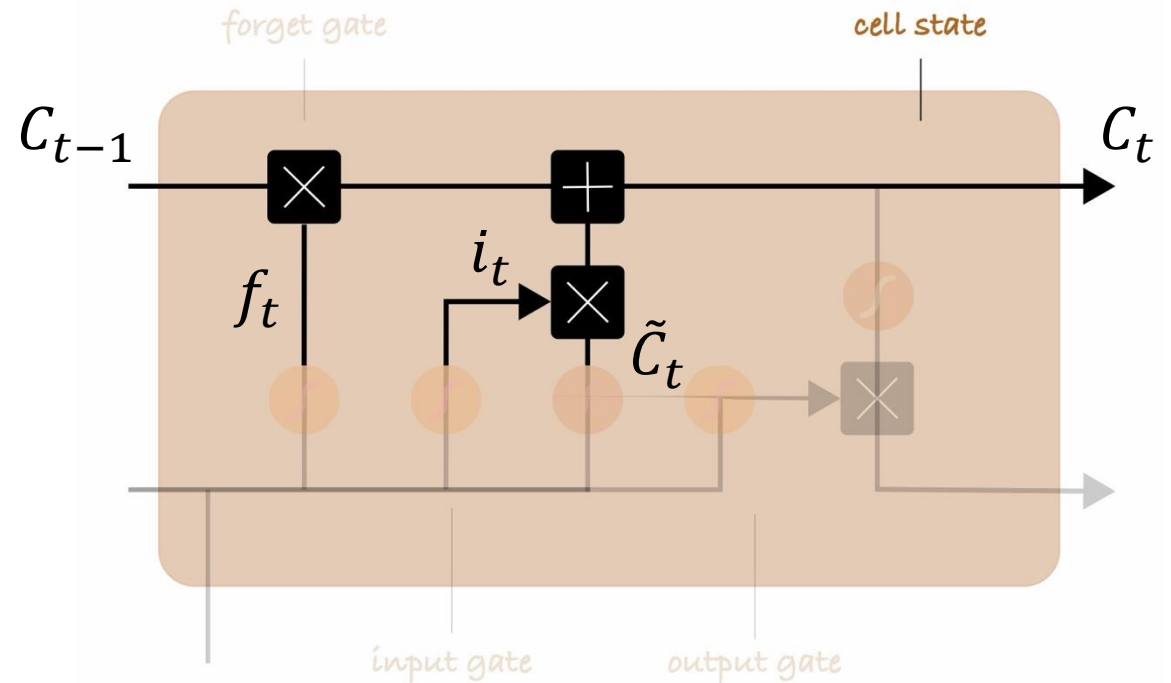
$$\tilde{C}_t = \tanh(W_c h_{t-1} + U_c x_t)$$



Step-by-Step LSTM Walkthrough

The **cell** state

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

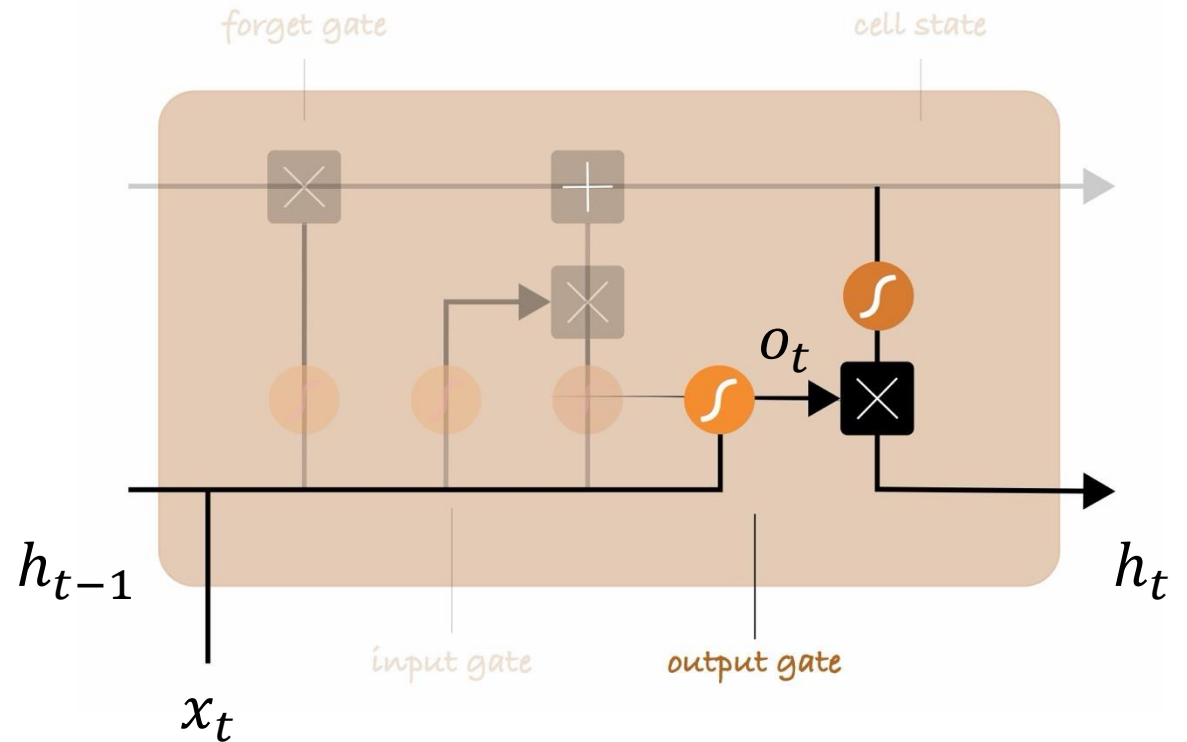


Step-by-Step LSTM Walkthrough

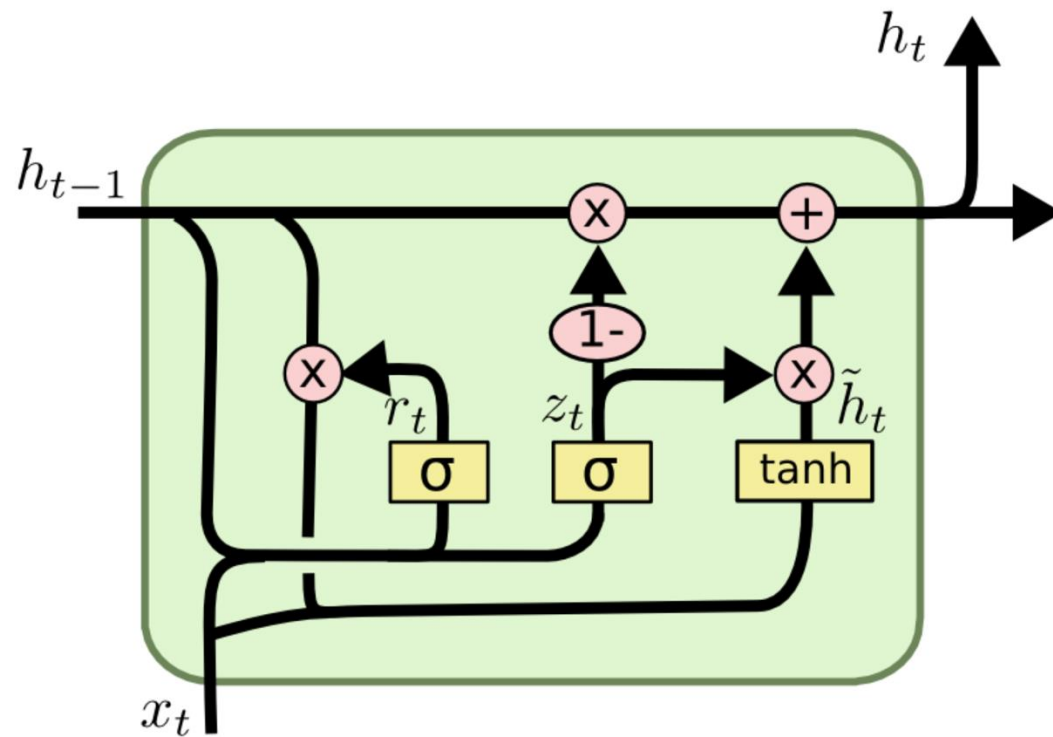
The **output** gate layer

$$o_t = \sigma(W_o h_{t-1} + U_o x_t)$$

$$h_t = o_t \tanh(C_t)$$



Gated Recurrent Units (GRU)



GRU equations

$$z_t = \sigma(W_z h_{t-1} + U_z x_t)$$

$$r_t = \sigma(W_r h_{t-1} + U_r x_t)$$

$$\tilde{h}_t = \tanh(W r_t h_{t-1} + U x_t)$$

$$h_t = (1 - z_t)h_{t-1} + z_t \tilde{h}_t$$



Orthogonal Units



Bounded Gradients

RNN derivatives

$$\frac{\partial L_T}{\partial h_t} = \frac{\partial L_T}{\partial h_T} \prod_{k=t}^{T-1} \frac{\partial h_{k+1}}{\partial h_k} = \frac{\partial L_T}{\partial h_T} \prod_{k=t}^{T-1} D_{k+1} W_k^T$$

Bounds

$$\begin{aligned} \left| \frac{\partial L_T}{\partial h_t} \right| &= \left| \frac{\partial L_T}{\partial h_T} \prod_{k=t}^{T-1} D_{k+1} W_k^T \right| \\ &\leq \left| \frac{\partial L_T}{\partial h_T} \right| \prod_{k=t}^{T-1} |D_{k+1} W_k^T| \\ &= \left| \frac{\partial L_T}{\partial h_T} \right| \prod_{k=t}^{T-1} |D_{k+1}| \end{aligned}$$



Orthogonal Matrices

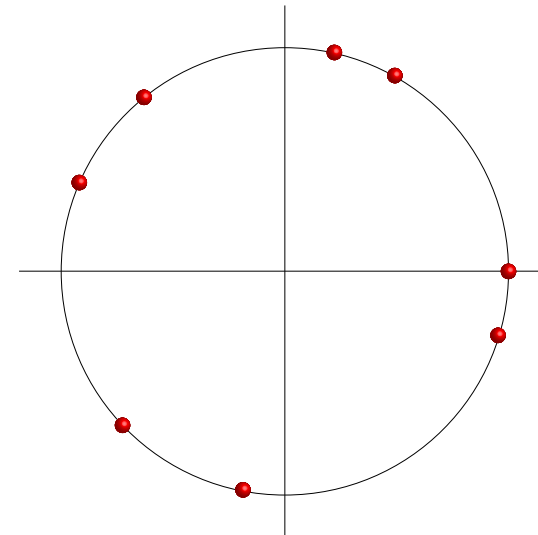
A real square W is **orthogonal** if

$$W^T W = I$$

$\uparrow$
 $W^{-1} = W^T$

We have

$$|\lambda_j| = 1, \quad \forall j$$



$$h_t = \sigma(b + Wh_{t-1} + Vx_t)$$

Orthogonal RNN

Force the hidden-to-hidden weights W to be orthogonal

$$W = D_3 R_2 F^{-1} D_2 \Pi R_1 F D_1$$

diagonal

reflection

transforms

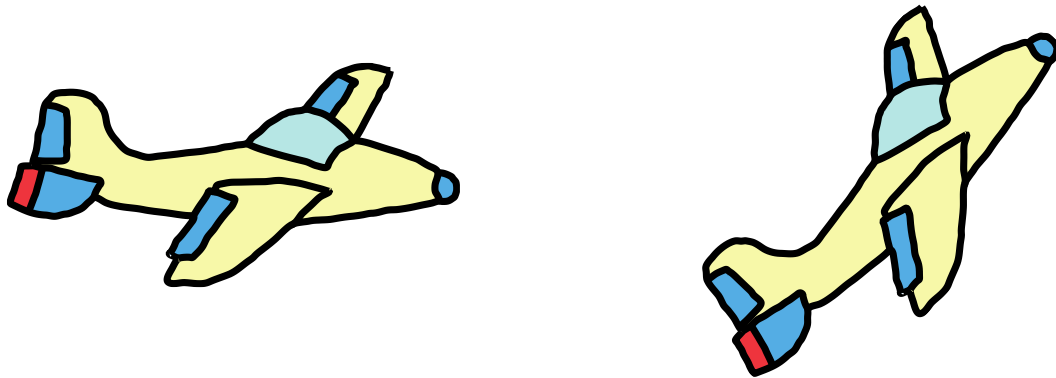
permutation



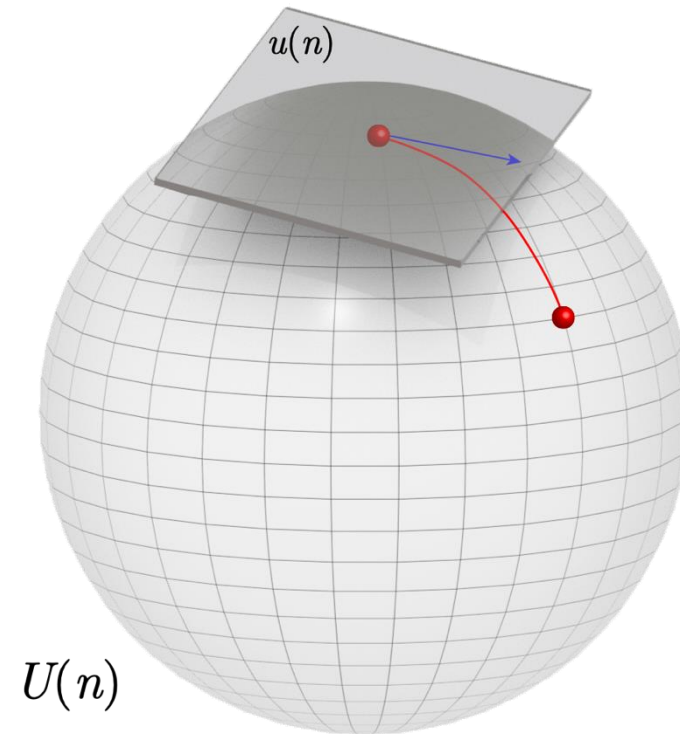
Orthogonal Matrices as Rotations

In 2D:

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



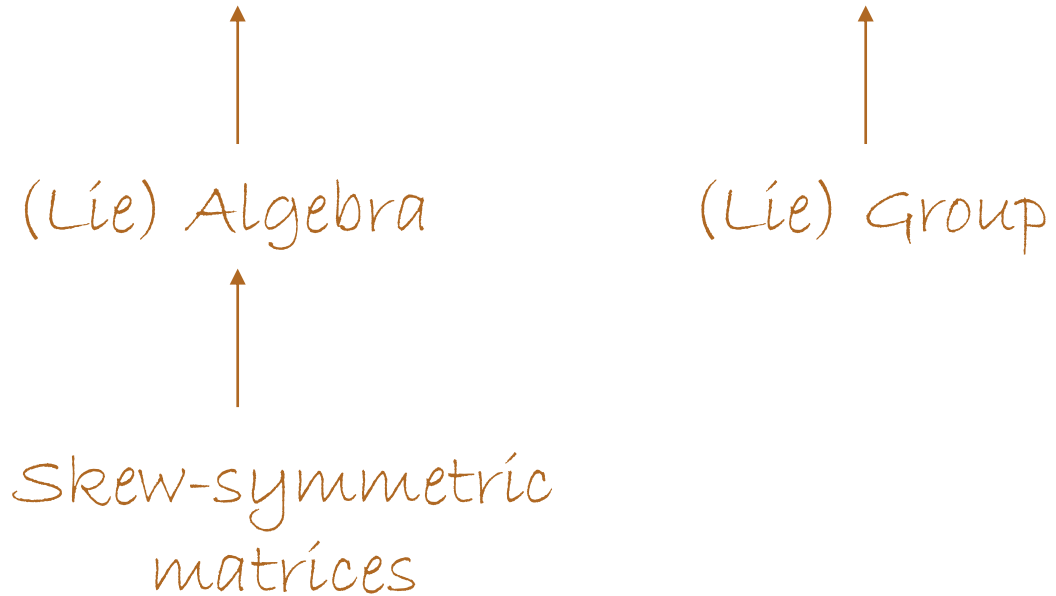
More generally:



$SO(n)$ and expRNN

Based on (Lie) Group Theory

$$A \in \mathfrak{so}(n) \Rightarrow \exp(A) \in SO(n)$$



The expRNN model

$$h_t = \sigma(\exp(A)h_{t-1} + Ux_t)$$

Diagram showing the input $A \in \mathfrak{so}(n)$ (written in orange) pointing up to the $\exp(A)$ term in the equation above.

$$A \in \mathfrak{so}(n)$$

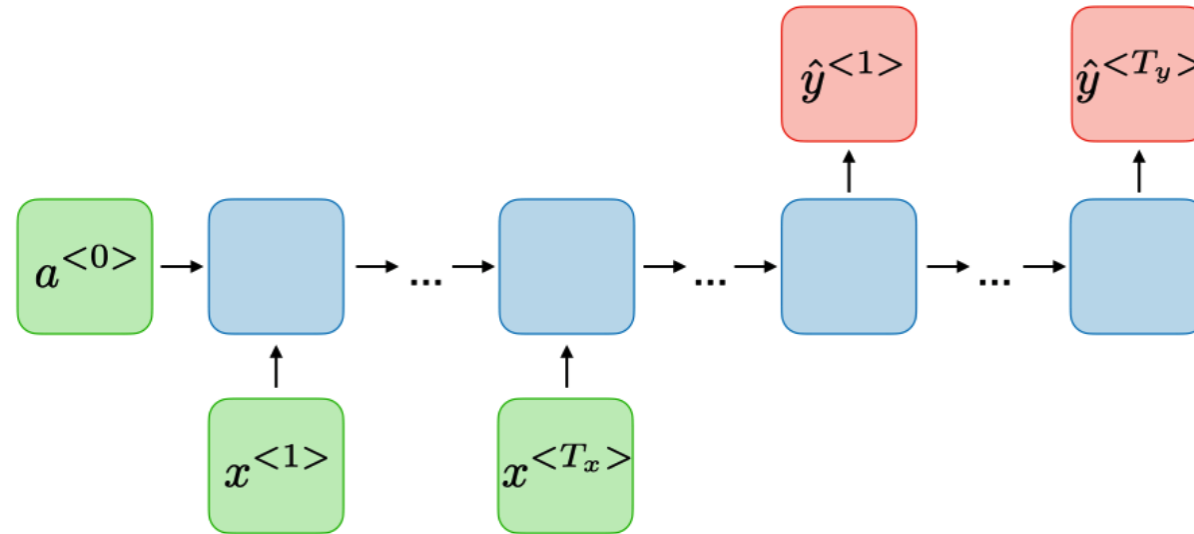


Sequence-to-Sequence



Design Patterns

Many-to-many
 $T_x \neq T_y$



Applications

Speech Recognition



Text Translation



Question Answering

Passage Sentence

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under gravity.

Question

What causes precipitation to fall?

Answer Candidate

gravity

- Between question and answer

cause---gravity

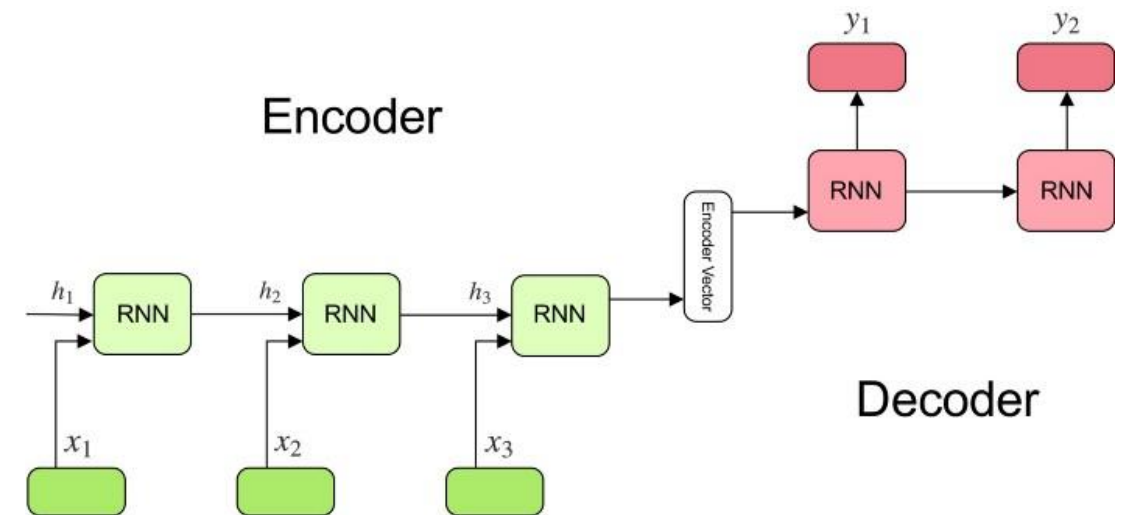
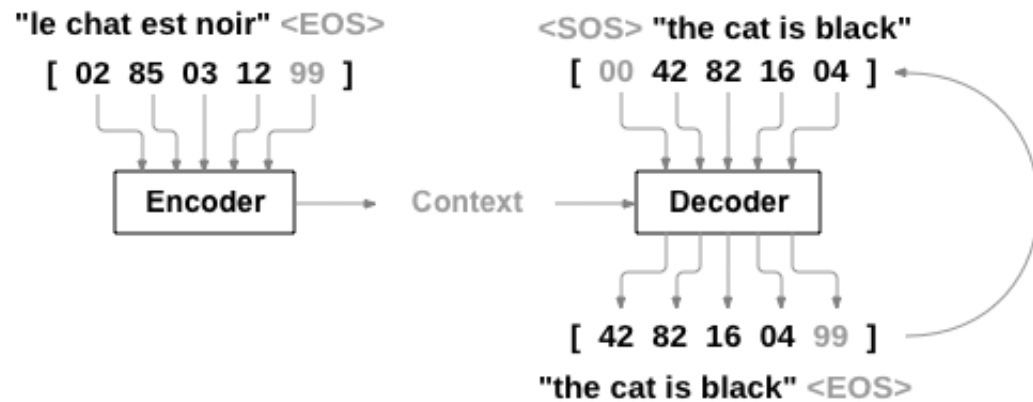
precipitation---gravity

fall---gravity

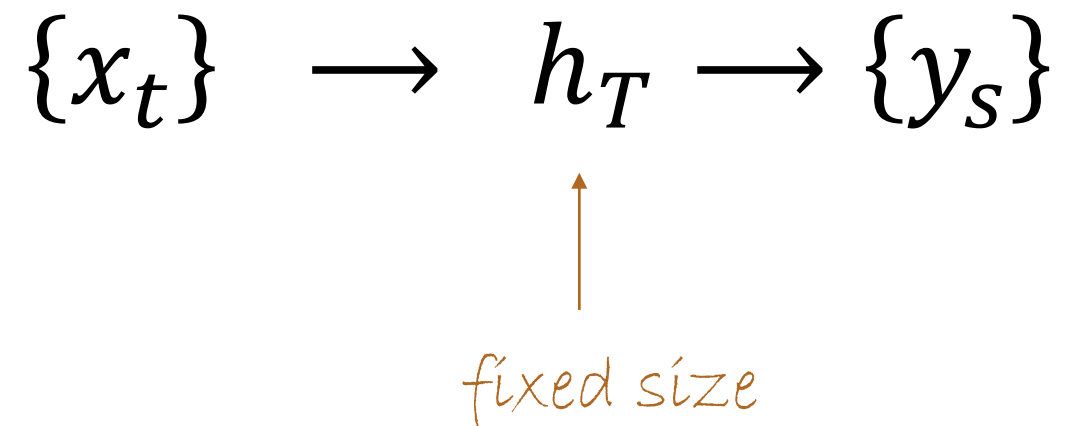
what---gravity



Seq2Seq Models



The Context Vector

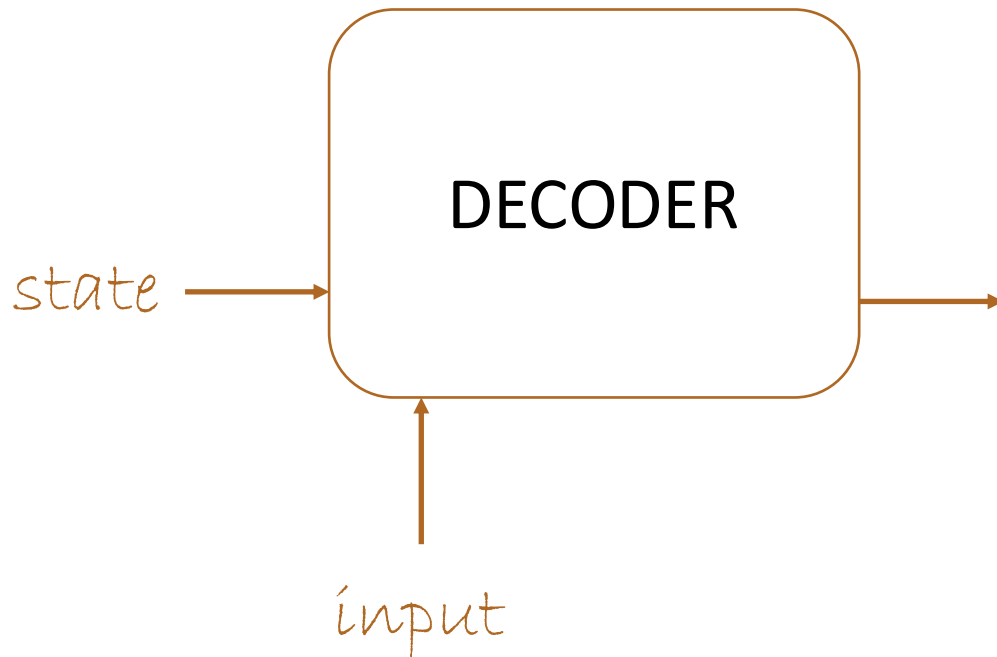


How to use the context

What is the input of the **decoder**?

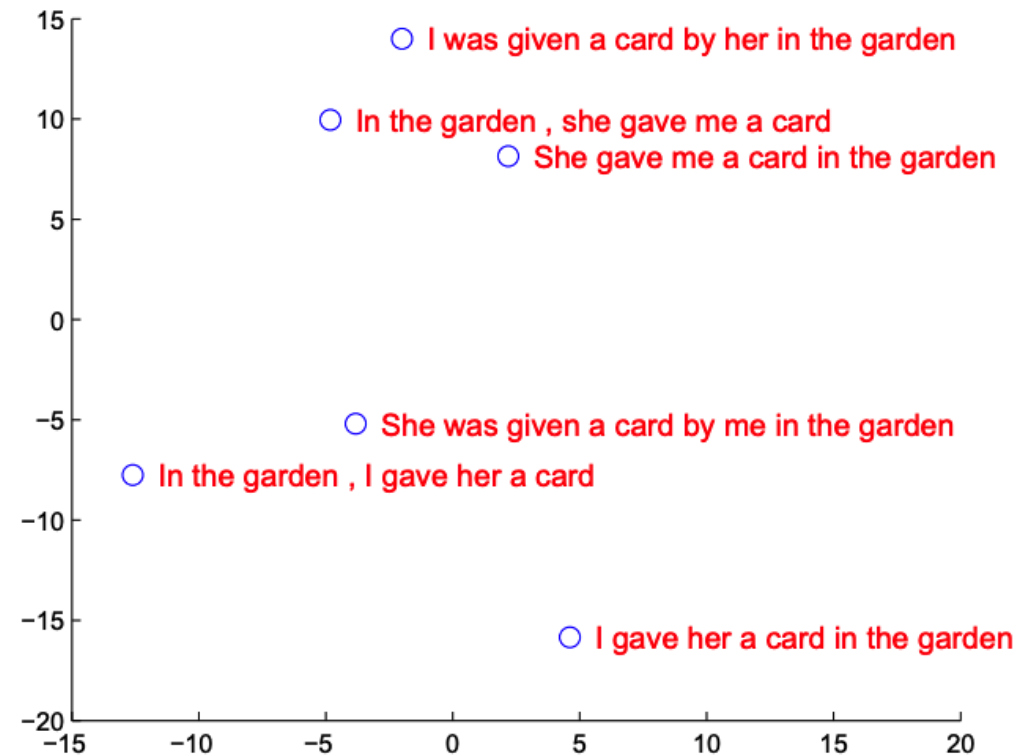
There are two main options:

1. always use the **context**
2. feed the temporal **output**



A Few Results

Type	Sentence
Our model	Ulrich UNK , membre du conseil d' administration du constructeur automobile Audi , affirme qu' il s' agit d' une pratique courante depuis des années pour que les téléphones portables puissent être collectés avant les réunions du conseil d' administration afin qu' ils ne soient pas utilisés comme appareils d' écoute à distance .
Truth	Ulrich Hackenberg , membre du conseil d' administration du constructeur automobile Audi , déclare que la collecte des téléphones portables avant les réunions du conseil , afin qu' ils ne puissent pas être utilisés comme appareils d' écoute à distance , est une pratique courante depuis des années .
Our model	“ Les téléphones cellulaires , qui sont vraiment une question , non seulement parce qu' ils pourraient potentiellement causer des interférences avec les appareils de navigation , mais nous savons , selon la FCC , qu' ils pourraient interférer avec les tours de téléphone cellulaire lorsqu' ils sont dans l' air ” , dit UNK .
Truth	“ Les téléphones portables sont véritablement un problème , non seulement parce qu' ils pourraient éventuellement créer des interférences avec les instruments de navigation , mais parce que nous savons , d' après la FCC , qu' ils pourraient perturber les antennes-relais de téléphonie mobile s' ils sont utilisés à bord ” , a déclaré Rosenker .



Deep RNN Models

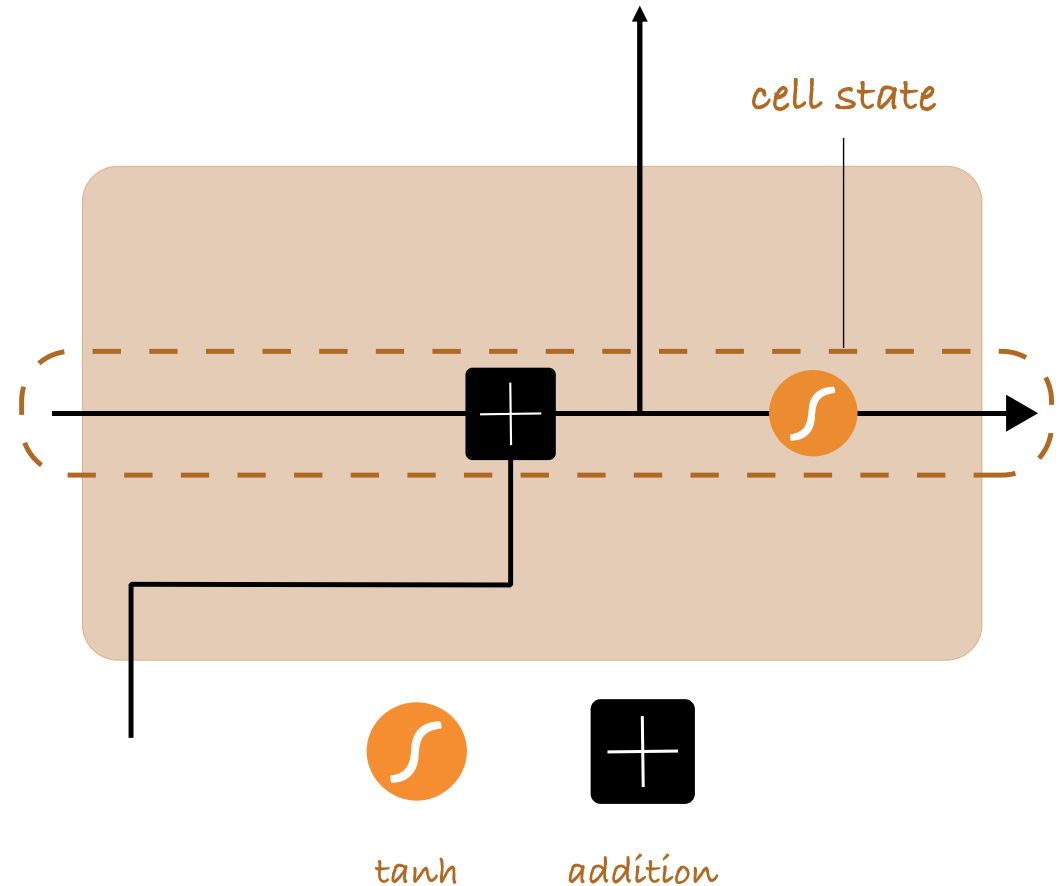


Vanilla RNN

RNN equations:

$$h_t = \sigma(W h_{t-1} + U x_t)$$

$$y_t = \varphi(V h_t)$$



Deep RNN?

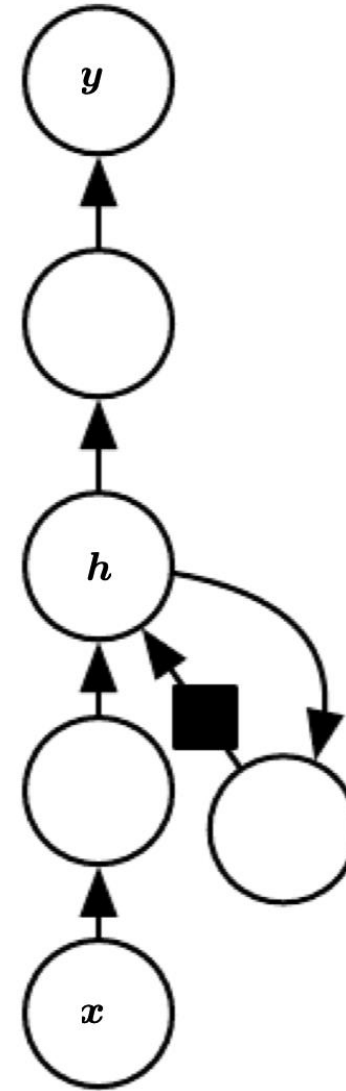
$$h_t = \sigma(F(\theta_F)h_{t-1} + G(\theta_G)x_t)$$

MLP

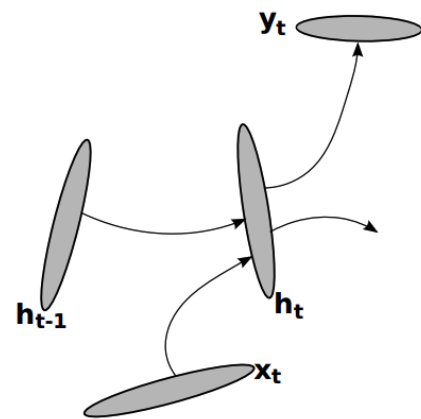
MLP

$$y_t = \varphi(H(\theta_H)h_t)$$

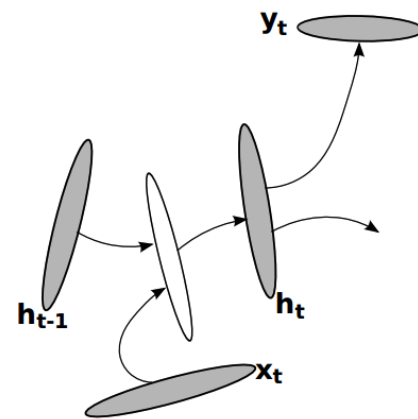
MLP



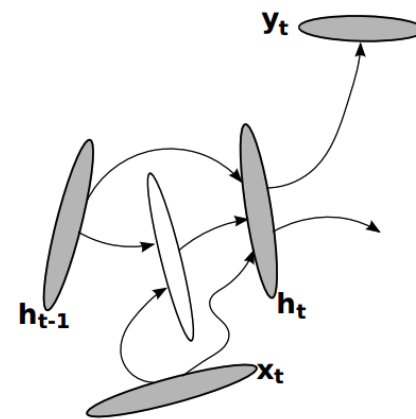
Deep Recurrent Models



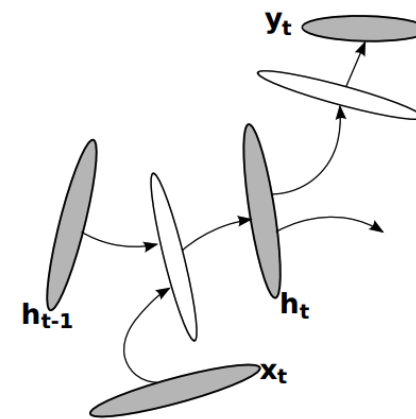
(a)



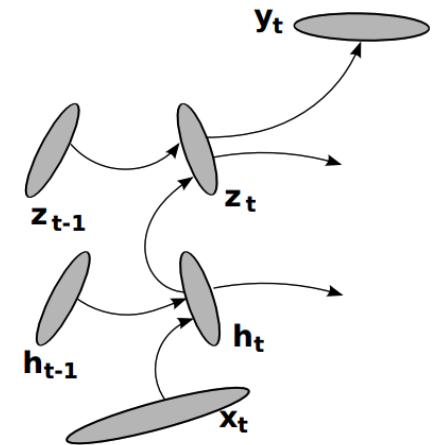
(b)



(c)



(d)



(e)

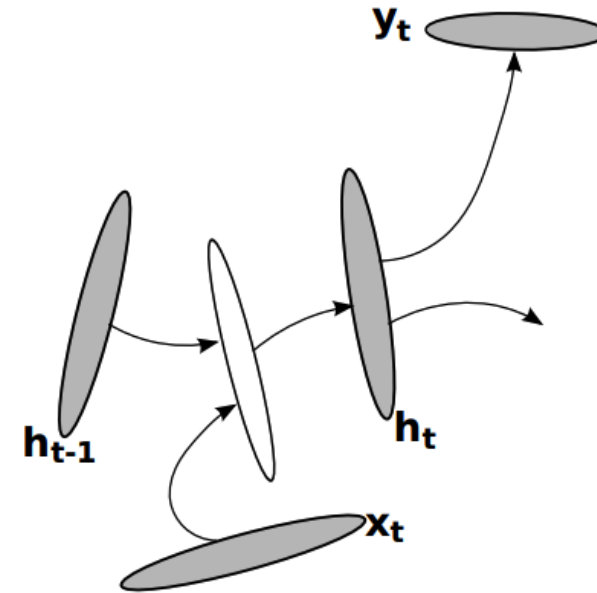


Deep Transition Models

Using L intermediate layers

$$\phi_h(W_L (\dots \phi_1(W_1 h_{t-1} + U x_t) \dots))$$

h_t

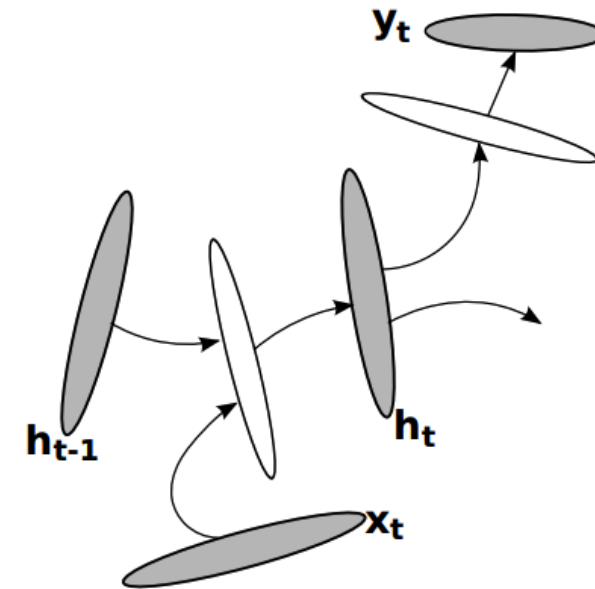


Deep Output Models

Using L intermediate layers

$$\phi_o(V_L (\dots \phi_1(V_1 h_t) \dots))$$

y_t



A Few Results

Shallow
RNN

	RNN	DT(S)-RNN	DOT(S)-RNN	sRNN	*	★
Character-Level	1.414	1.409	1.386	1.412	1.41 ¹	1.24 ³
Word-Level	117.7	112.0	107.5	110.0	123 ²	117 ³

LSTM



Bidirectional RNN



Using Past and Future Inputs

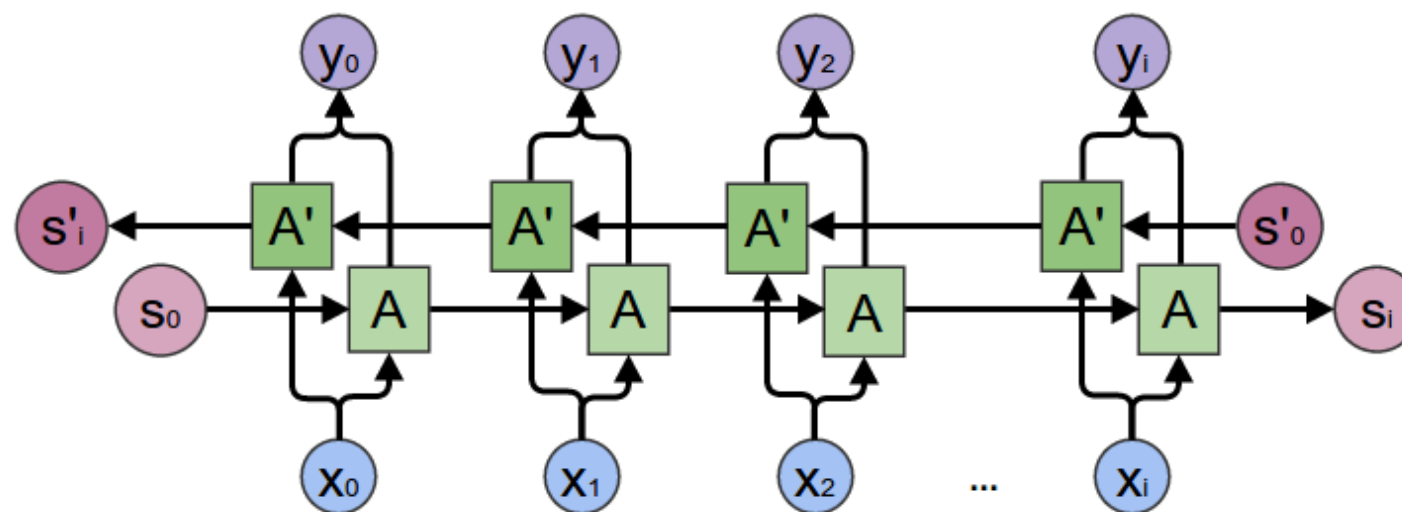
I am ____.

I am ____ hungry.

I am ____ hungry, and I can eat half a pig.



Bidirectional RNN



Forward and Backward States

The **output** is typically generated by

$$y_i = \sigma(W[s_i, s'_0] + b)$$



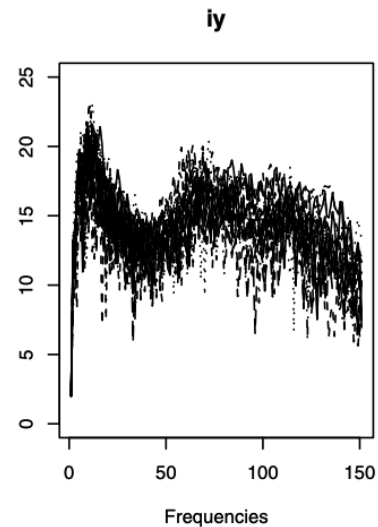
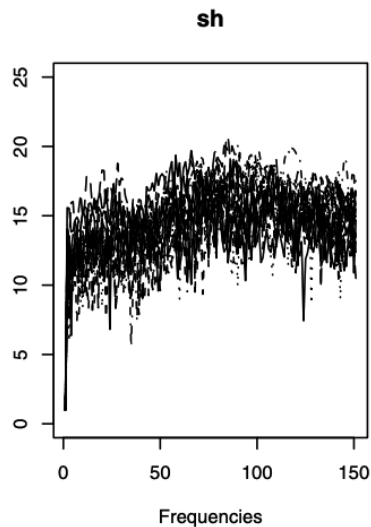
concatenation



Results on Speech Recognition

TIMIT dataset

log-periodograms



System	Training Set	Test Set
Bidirectional LSTM	78.1%	73.2%
Unidirectional LSTM	75.2%	70.1%
Bidirectional RNN	66.6%	65.3%
Unidirectional RNN	63.0%	61.9%

